TU/CSE

TEZPUR UNIVERSITY

Spring Semester End Term Examination, 2022

CO 315 (B.Tech): COMPUTER NETWORKS

Full Marks: 60

Time: 120 mins

Figures in the right hand indicate marks for the individual question.

1 Answer any seven of the followings: 7x2

i. What is the role of Fiestel Cipher in DES?

ii. What is ICMP redirect/change request? Who initiates the redirect request?

iii. Working of NAT (Network Address Translation).

iv. How does ICMP handle congestion?

v. How does the Diffie-Hellmen Protocol handle man-in-the-middle attack?

vi. What are the basic BGP messages?

vii. Differential services for L3 QoS.

viii. How do you distinguish IMAP and SMTP? How do they co-operate while sending an email?

ix. Differentiate between Autonomous Systems and OSPF Areas.

2 Information to be transmitted over the internet contains the following characters with their associated frequencies (shown in parenthesis): 8

a(45)e(65)l(13)n(45)o(18)s(22)t(53)

Use Huffman technique to answer the following questions:

A. Build the Huffman code tree for the message, B. Use the Huffman tree to find the codeword for each character, C. If the data consists of only these characters, what is the total number of bits to be transmitted? D. What is the compression ratio? E. Verify that your computed Huffman codewords satisfy the Prefix property.

3 Name the three fields that are added in RIPv2 header as compared to that of RIPv1. Illustrate count-to-infinity problem for RIP. Name a method to solve the problem. How does OSPF perform Neighbour Discovery and Database Synchronization? Mention the types of restricted areas in OSPF, and differentiate them. 3+2+1+4+4

4 Mention any four functions of Proxy/Gateway agent in SNMP. Describe the four basic operations of SNMP nodes with a single sentence for each. 4+4

5 Mention the path selection decision attributes of BGP in terms of LOCAL-PREF and AS-PATH. Mention any two advantages of path vector routing over distance vector routing. 2+2

7 Describe the UDP header fields with a neat diagram. What do you understand by the conditions when in a UDP datagram: i. Checksum value received is all 0s, ii. Sender includes checksum but the value of the sum is all 0s. 4+2

8 How does the traffic throttling with feedback mechanism work for congestion control? Provide an example of the same and also show how feedback mechanism works in it. Differentiate between leaky bucket and token bucket traffic shaping algorithms. 2+2+2

Mid-term, Spring Semester, 2022
CO 315 (B.Tech), CSE, Tezpur University,
Date: 28/03/22, FM: 40, FT: 120 mins

1 The routing table of a router is shown below: 2+2

Destination	Sub-net mask	Interface
128.75.43.0	*255.255.255.0*	*Eth0*
128.75.43.0	*255.255.255.128*	*Eth1*
192.12.17.5	*255.255.255.255*	*Eth3*
default		*Eth2*

If no match occurs, the option default is chosen.
On which interfaces will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively and why? Explain your answers.

2 Which statements describe the IP address 10.16.3.65/23? Justify your answer. 1+
1. The subnet address is 10.16.3.0 255.255.254.0.
2. The lowest host address in the subnet is 10.16.2.1 255.255.254.0.
3. The last valid host address in the subnet is 10.16.2.254 255.255.254.0.
4. The broadcast address of the subnet is 10.16.3.255 255.255.254.0.

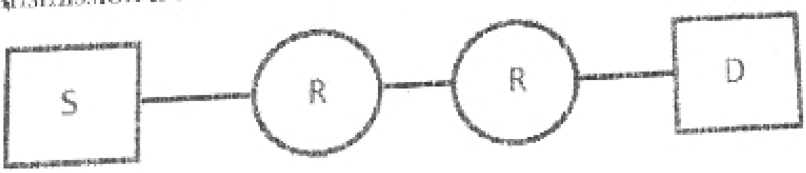
3 Describe the significance of ToS in IP? What is the operation of TTL?

4 An unfragmented packet of 10000 bytes is communicated. At the arrived routers, fragmentation is done in two levels. In first level, four fragments of 2500 bytes are formed. In the second level, further fragmentation is done to forward the packet to an Ethernet network. Show the fragments offsets and fragmentation flags of each fragment.

5 A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet? Justify your answer.

6 A company has a class C network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Provide a set of three valid subnets to support the same? Justify your answer,

7 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D. Justify and explain properly.



8 Provide socket system calls that support TCP connection set-up, data transfer and connection termination. Show the TCP messages communicated in each step. Also show the variations in the sequence numbers, acknowledgement number and flags. How is the ICN chosen?